# Wi-Fi®/Bluetooth® connectivity expansion board based on the ST67W611M1 module for STM32 Nucleo boards

## Introduction

The X-NUCLEO-67W61M1 expansion board provides an affordable and flexible way for users to try out new concepts and build prototypes with the ST67W611M1 coprocessor module.

Interfacing with the host is possible through SPI using the ARDUINO® Uno V3 or expansion connectors. This expansion board is stackable on Nucleo-64, Nucleo-144, and Discovery boards. An ST67W611M1A6U module with an embedded RF connector can also be implemented on the X-NUCLEO-67W61M1 expansion board.

Powered by all-in-one 2.4 GHz Wi-Fi® 6/Bluetooth® combo SoC, the ST67W611M1 LGA module is purposely designed to pack the processing capabilities, Wi-Fi®/Bluetooth® combo connectivity, and on-module memory into a single 32-pin LGA form factor.

For more details on the ST67W611M1 coprocessor module, refer to its datasheet at *www.st.com*.

**Figure 1. X-NUCLEO-67W61M1 top view**

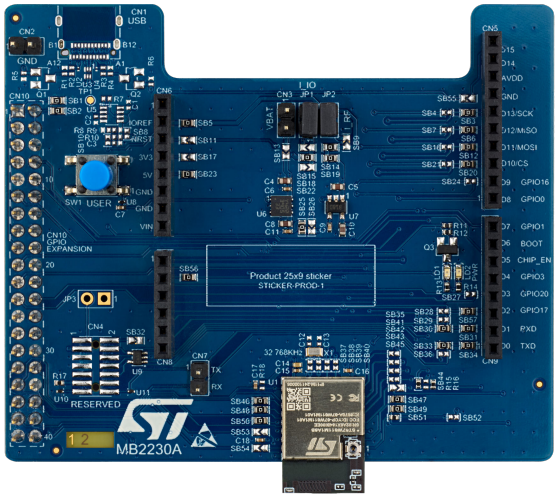
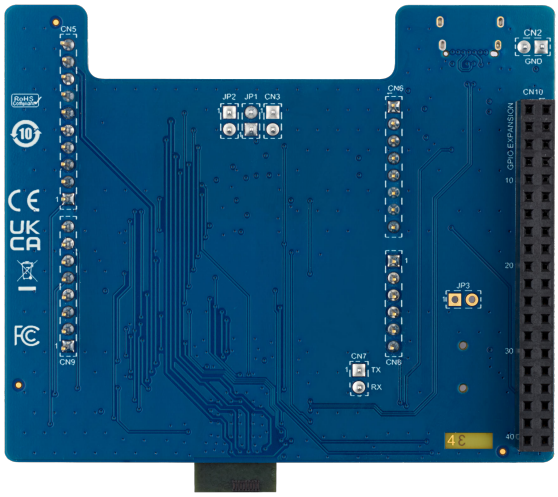



**Figure 2. X-NUCLEO-67W61M1 bottom view**



*Pictures are not contractual.*

# 1 Features

- ST67W611M1 module, 32-pin, 4-side LGA 1.27-mm pitch (12.28 x 17.28 x 2.2 mm) package featuring:
  - 2.4 GHz Wi-Fi®/Bluetooth® combo all-in-one SoC
  - Wi-Fi® 6, coprocessor IEEE 802.11 b/g/n/ax
  - Single-band 2.4 GHz
  - Low-power Wi-Fi® with various sleep modes
  - PCB antenna and antenna connector
- User LED
- User push button
- Board connectors:
  - ARDUINO® Uno V3
  - Raspberry Pi® 40-pin GPIO header
- Scalable solution, capable of cascading multiple boards for larger systems
- Free comprehensive development firmware library and examples, compatible with the X-CUBE-ST67W61 expansion software package for STM32Cube

# 2 Ordering information

To order the X-NUCLEO-67W61M1 expansion board, refer to Table 1. Additional information is available from the datasheet and reference manual of the ST67W611M1 module at *www.st.com*.

**Table 1. Ordering information**

| Order code | Board reference | Target ST67 |
|---|---|---|
| X-NUCLEO-67W61M1 | MB2230(1) | ST67W611M1A6B |

1. *Expansion board*

## 2.1 Codification

The meaning of the codification is explained in Table 2.

**Table 2. Codification explanation**

| X-NUCLEO-XXWYZMT | Description | Example: X-NUCLEO-67W61M1 |
|---|---|---|
| X-NUCLEO | Type of board | STM32 Nucleo expansion boards |
| XX | MCU series in coprocessor series | ST67W ST Wi-Fi® coprocessor series |
| W | Wireless technology | Wi-Fi® |
| Y | Protocol | Wi-Fi® protocol: Wi-Fi® 6 |
| Z | Die version | Die version 1 |
| M | Module | Module |
| T | Sequential number | First Wi-Fi® Nucleo expansion board |

# 3 Development environment

## 3.1 EDA resources

All board design resources, including schematics, EDA databases, manufacturing files, and the bill of materials, are available from the X-NUCLEO-67W61M1 product page at *www.st.com*.

# 4 Conventions

Table 3 provides the conventions used for the ON and OFF settings in the present document.

**Table 3. ON/OFF convention**

| Convention | Definition |
|---|---|
| Jumper JPx ON | Jumper fitted |
| Jumper JPx OFF | Jumper not fitted |
| Jumper JPx [1-2] | Jumper fitted between pin 1 and pin 2 |
| Solder bridge SBx ON | SBx connections closed by 0 Ω resistor |
| Solder bridge SBx OFF | SBx connections left open |
| Resistor Rx ON | Resistor soldered |
| Resistor Rx OFF | Resistor not soldered |
| Capacitor Cx ON | Capacitor soldered |
| Capacitor Cx OFF | Capacitor not soldered |

# 5 Safety recommendations

## 5.1 Targeted audience

This product targets users with at least basic electronics or embedded software development knowledge like engineers, technicians, or students.

This board is not a toy and is not suited for use by children.

## 5.2 Handling the board

This product contains a bare printed circuit board and as with all products of this type, the user must be careful about the following points:

- The connection pins on the board might be sharp. Be careful when handling the board to avoid hurting yourself
- This board contains static-sensitive devices. To avoid damaging it, handle the board in an ESD-proof environment.
- While powered, do not touch the electric connections on the board with your fingers or anything conductive. The board operates at voltage levels that are not dangerous, but components might be damaged when shorted.
- Do not put any liquid on the board. Avoid operating the board close to water or at a high humidity level.
- Do not operate the board if dirty or dusty.

## 5.3 Delivery recommendations

Before the first use, inspect the board for any damage that may have occurred during shipment. Ensure that all socketed components are securely fixed in their sockets and that nothing is loose in the plastic bag.

## 5.4 Power supply

A power supply unit or auxiliary equipment complying with the EN 62368-1:2014+A11:2017 standard (or the one replacing it) and safety extralow voltage (SELV/ES1) with limited power capability (LPS/PS2) must power this equipment.

# 6 Quick start

The X-NUCLEO-67W61M1 expansion board product is an easy-to-use and low-cost development kit used to evaluate and start development quickly with an ST67W611M1A6B module. Before installing and using the product, accept the evaluation product license agreement from the *www.st.com/epla* web page. For more information on the X-NUCLEO-67W61M1 expansion board product and demonstration software, visit the *www.st.com/stm32nucleo* web page.

## 6.1 Getting started

Follow the sequence below to configure the product and launch the demonstration application (refer to Figure 4 for component location):

1. Check jumper positions on board, JP1 (IDD_RF) ON, JP2 (IDD_IO) ON, JP3 (internal) OFF. The default jumper position on the board is explained in Table 4.
2. Connect the X-NUCLEO-67W61M1 board to a host (such as a NUCLEO-U575ZI board) through the ARDUINO® connector.
3. Connect the host board to a PC using the USB connector (CN1 on NUCLEO-U575ZI) to a PC to supply power to the board and establish a connection with ST-LINK.
4. Launch *NCP_update_mission_profile_t01.bat*. For further information, refer to user manual *Getting started with X-CUBE-ST67W61* (UM3475).
5. On the PC, connect a UART terminal, such as TeraTerm, to the board using the following settings:
   – UART terminal: select new line received = LF; new line transmit = LF (line feed).
   – Serial port setting: Select COM port number, 921600 baud rate, 8-bit data, parity none, one stop bit, and no flow control.
6. Press the black RESET button of the host board, and press “enter” on a UART console.
7. Then enter “wifi_scan” to get all available Wi-Fi® networks.
8. This CLI application software is available on the *www.st.com* website.

*Note: For more information, refer to the corresponding wiki page on http://www.st.com.*

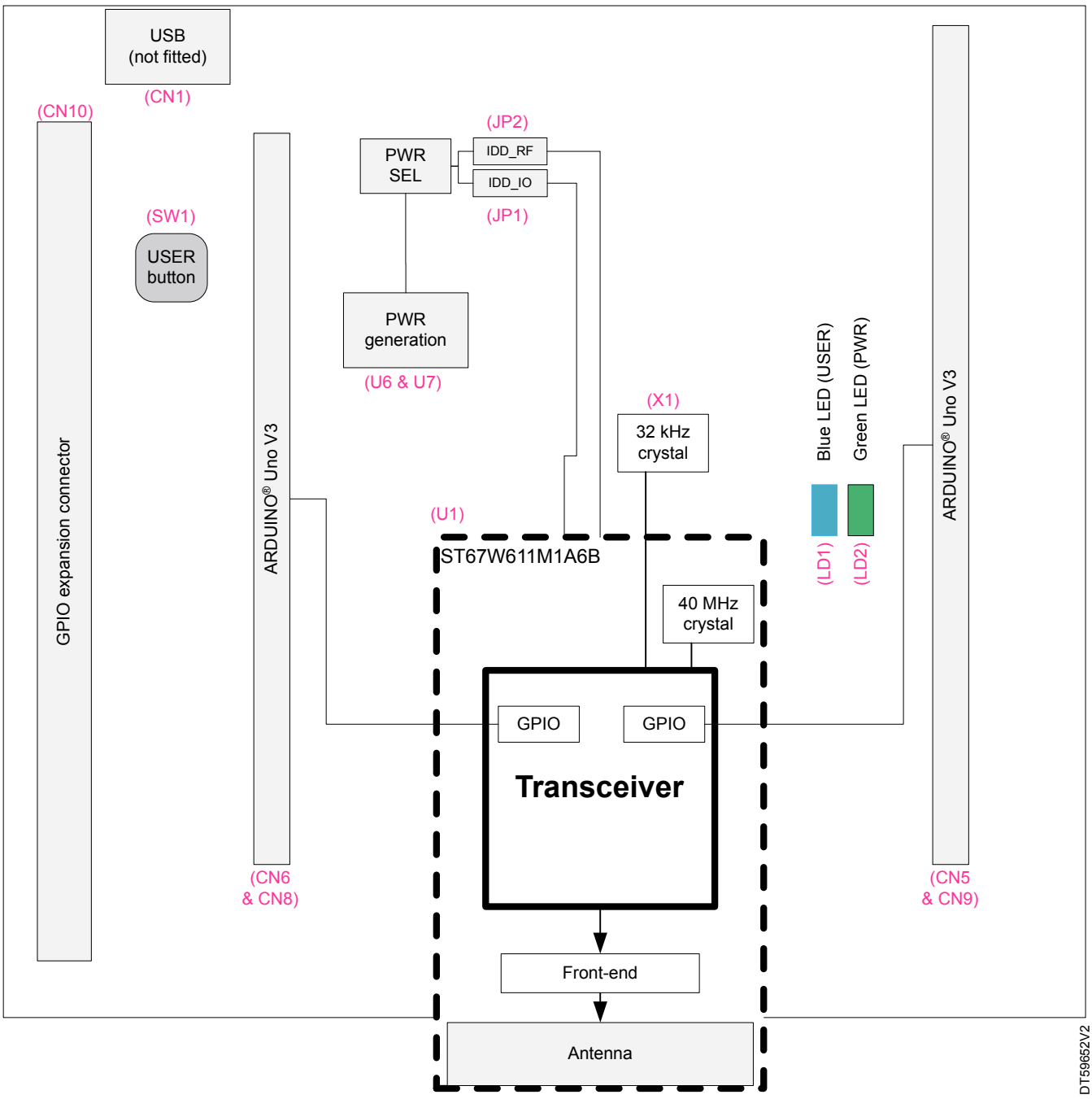
**Table 4. Jumper default configuration**

| Jumper | Definition | Position | Comment |
|---|---|---|---|
| JP1 | VDD_IO current measurement | ON | For ST67W611M1A6B current measurements (IO part) |
| JP2 | VDD_RF current measurement | ON | For ST67W611M1A6B current measurements (RF part) |
| JP3 | RESERVED | NA | JP3 not implemented |

# 7 Hardware layout and configuration

The X-NUCLEO-67W61M1 expansion board product is designed around the ST67W611M1A6B coprocessor module. Figure 3 shows the connections between the module and its peripherals such as push buttons, LEDs, or ARDUINO® Uno and GPIO expansion connectors. Figure 4 and Figure 5 show the location of these features on the X-NUCLEO-67W61M1 expansion board. The mechanical dimensions of the board are shown in Figure 6.

**Figure 3. X-NUCLEO-67W61M1 hardware block diagram**



## 7.1 PCB layout

Figure 4. **X-NUCLEO-67W61M1 top layout**

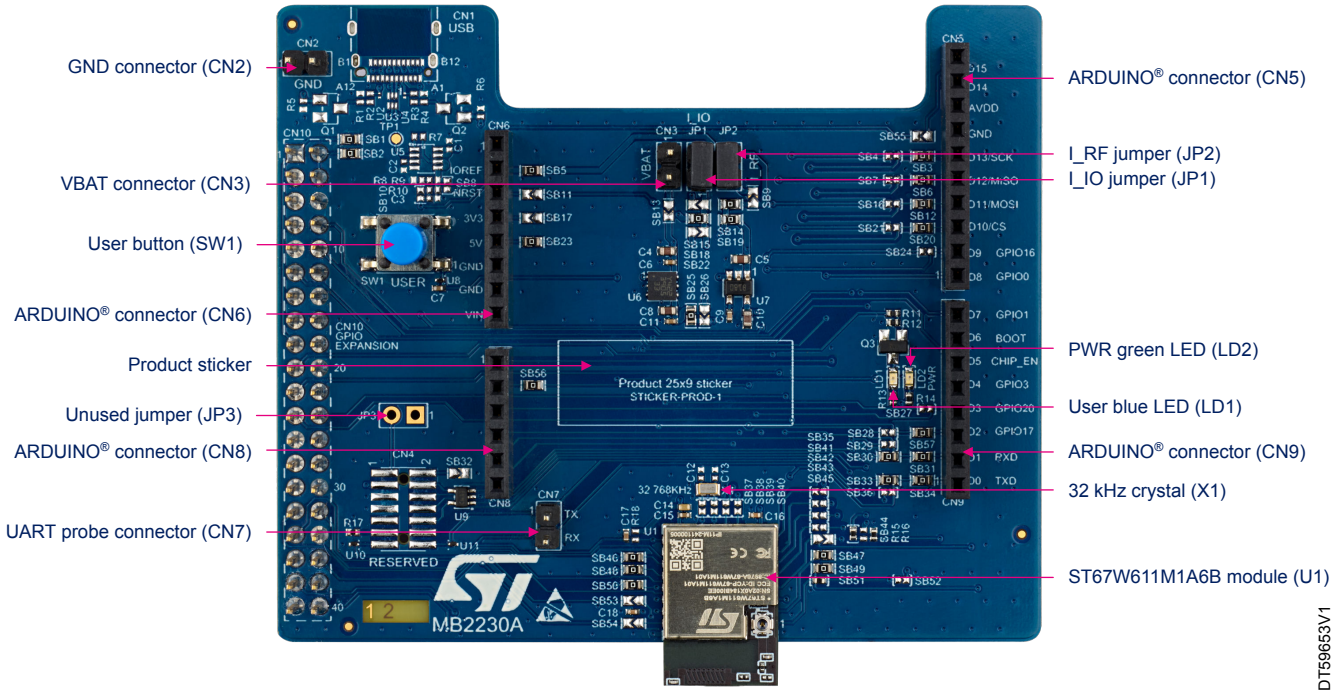
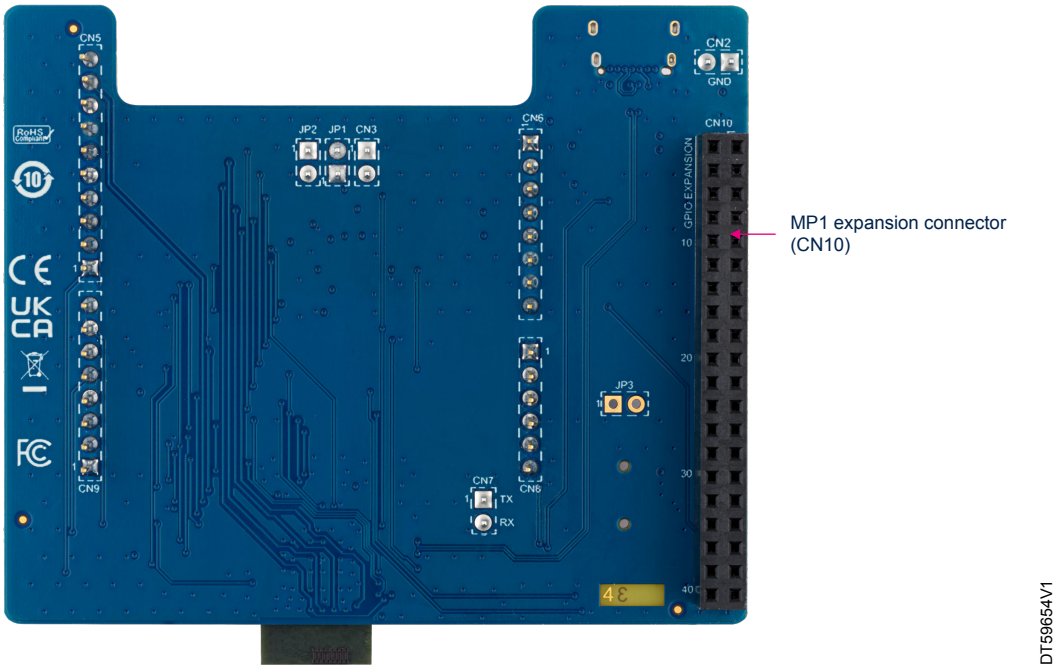
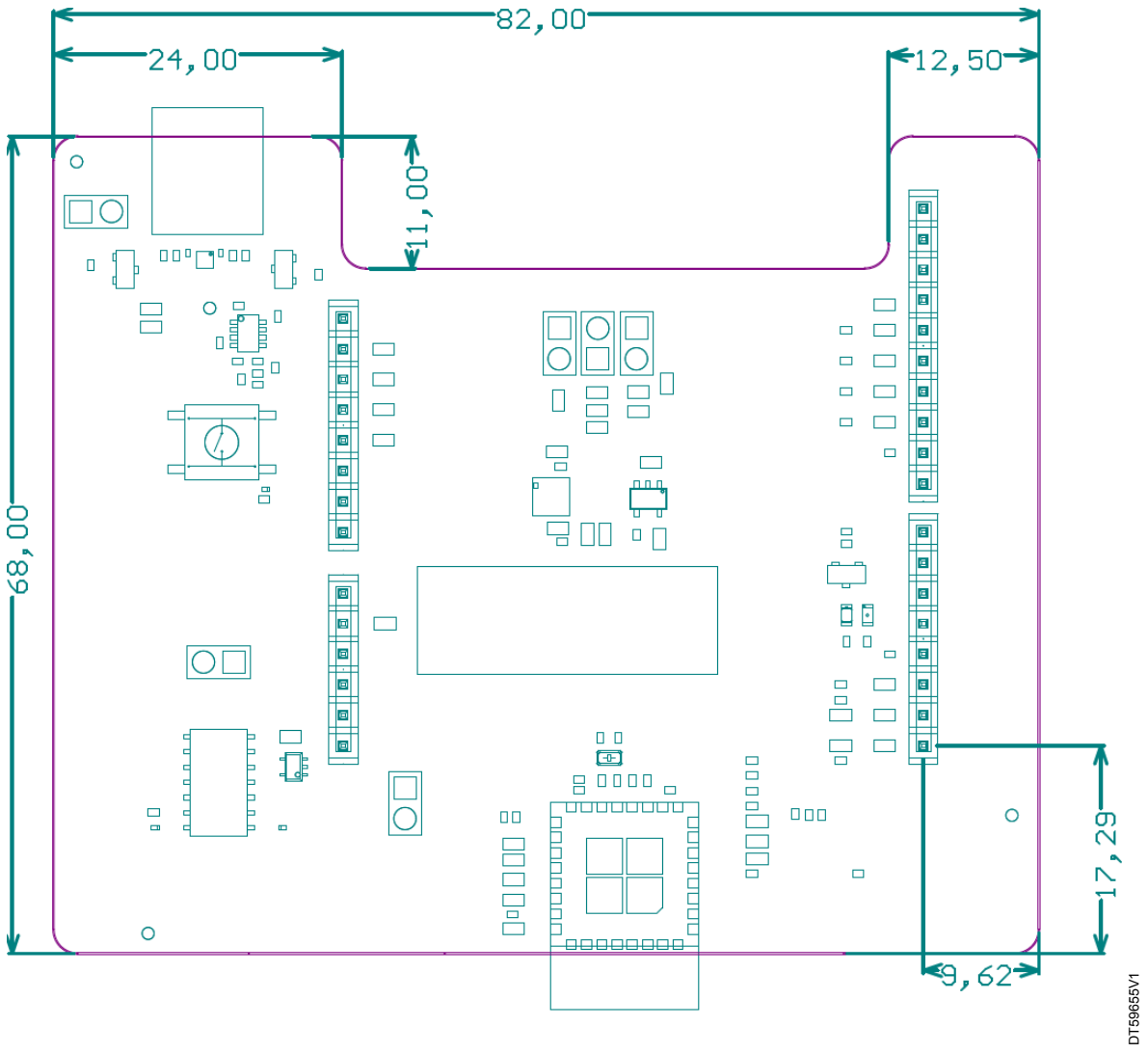



Figure 5. **X-NUCLEO-67W61M1 bottom layout**



## 7.2 Mechanical drawing

Figure 6. **X-NUCLEO-67W61M1 board mechanical drawing (in millimeters)**

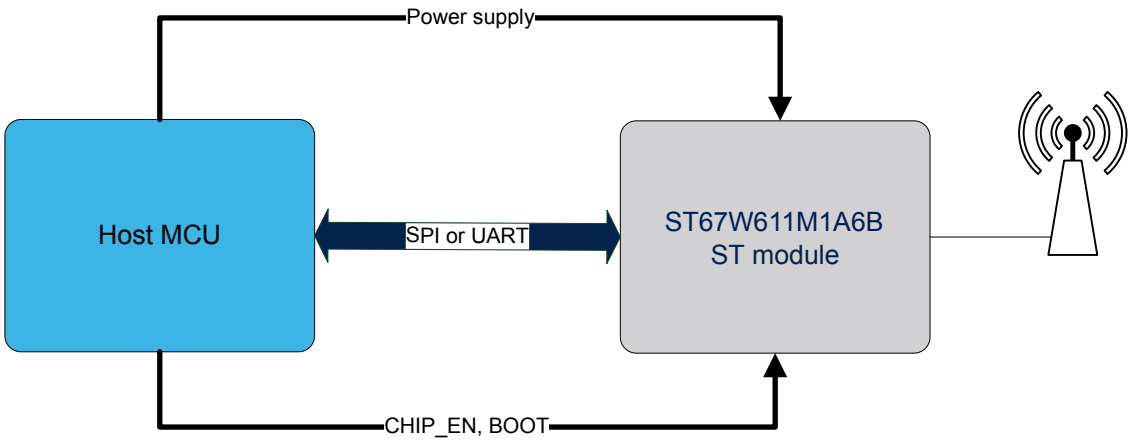


## 7.3 Board setup

The ST67W611M1A6B module is a coprocessor. Therefore, a host board with an MCU is required to manage communication and configuration. Figure 7 represents the interaction between the coprocessor module and its host.

Data exchange and configuration are performed through SPI and UART. CHIP_EN and BOOT signals are used to place the coprocessor in different modes.

**Figure 7. ST67W611M1A6B host interaction**




### 7.3.1 Main host boards and associated connectors

Different connectors might be used to interface X-NUCLEO with the host board.

In global cases, interfacing X-NUCLEO and STM32 boards is performed through:

- ARDUINO® for Nucleo-144 or Nucleo-64 boards
- Raspberry Pi® 40-pin GPIO header

**Table 5. X-NUCLEO host board connections**

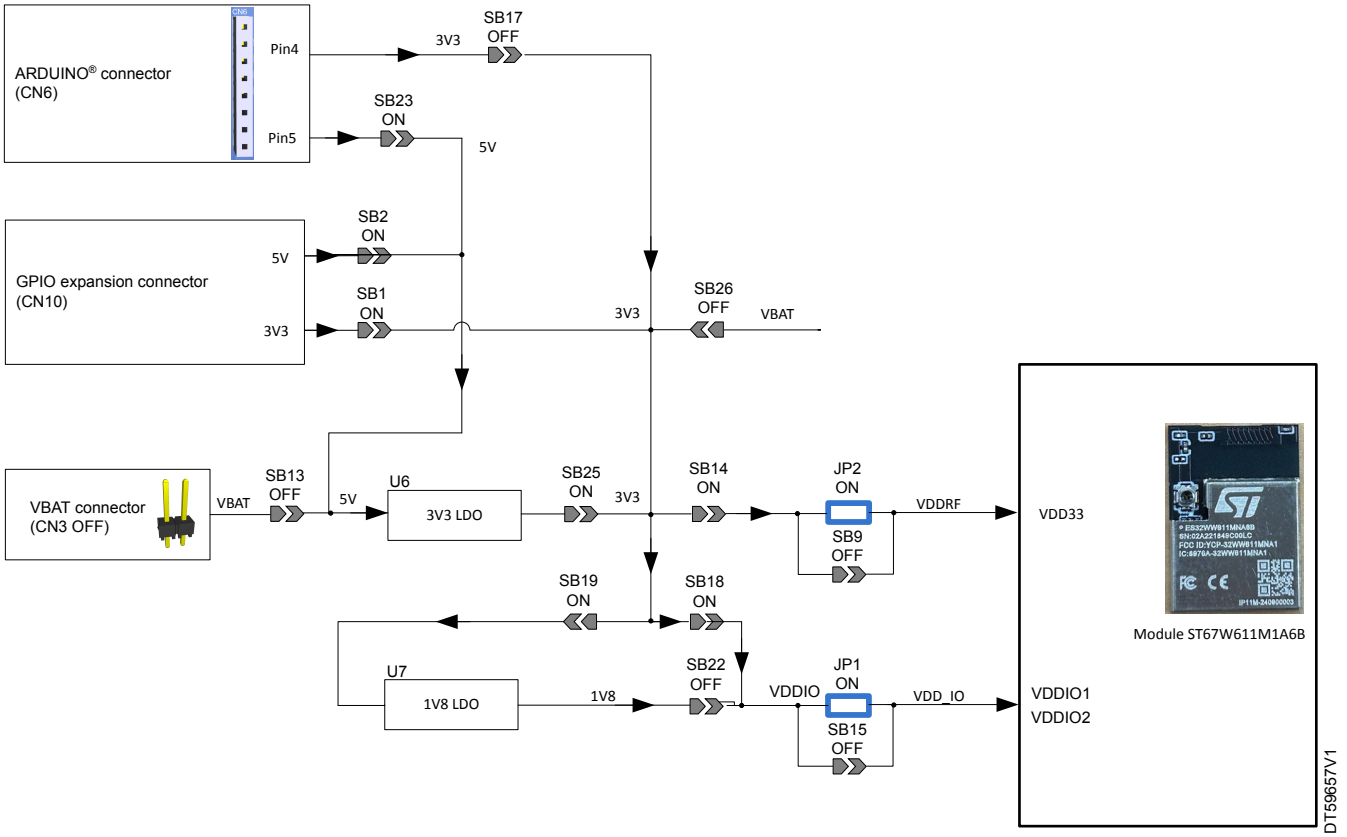
| Board reference | Description | Main interface | Connector used |
|---|---|---|---|
| MB1549 (NUCLEO-U575ZI-Q) | Nucleo-144 with STM32U575ZI-Q MCU | ARDUINO® Uno interface | CN6, CN5, CN8, and CN9 |
| MB1404 (NUCLEO-H563ZI) | Nucleo-144 with STM32H563ZI MCU | ARDUINO® Uno interface | CN6, CN5, CN8, and CN9 |
| MB1737 (NUCLEO-H7S3L8) | Nucleo-144 with STM32H7S3L8 MCU | ARDUINO® Uno interface | CN6, CN5, CN8, and CN9 |
| MB1940 (NUCLEO-N657X0-Q) | Nucleo-144 with STM32N657X0 MCU | ARDUINO® Uno interface | CN6, CN5, CN8, and CN9 |
| MB1272 (STM32MP157D-DK1 and STM32MP157F-DK2) | Discovery kit with STM32MP157D/F MPU | Raspberry Pi® 40-pin GPIO header | CN10 |

## 7.4 Power supply

The X-NUCLEO-67W61M1 expansion board offers various power supply configurations, depending on the used host board. In addition, the ST67W611M1A6B module has two power domains (VDD33 and VDDIO). The X-NUCLEO-67W61M1 board embeds 3.3 and 1.8 V LDO regulators. The user chooses the configuration by setting different solder bridges. Figure 8 shows the global power. The board supply and module domains are detailed in the next sections.

*Note: The power supply tree includes the board and ST67W611M1A6B module power domain supply selections, LDO, headers, and current measurement jumpers.*

**Figure 8. X-NUCLEO-67W61M1 power tree**



### 7.4.1 Board power supply

Different sources can provide the power supply:

- Supply 3V3 from the host through the ARDUINO® connector
- Supply 5V from the host through the ARDUINO® connector (default setting)
- Supply 3V3 from the host through the expansion connector
- Supply 5V from the host through the expansion connector
- An external battery cell or voltage generator connected to CN3 to provide 5V
- An external battery cell or voltage generator connected to CN3 to provide 3V3

For power supply selections, refer to Table 6 and the following sections.

**Table 6. Supply source selection**

| Main supply source(1) | Connector(1) | SB ON(1) | SB OFF(1) |
|---|---|---|---|
| ARDUINO® 3.3 V | CN6 | SB17 | SB13, SB23, SB25, and SB26 |
| **ARDUINO® 5 V** | **CN6** | **SB23 and SB25** | **SB13, SB17, and SB26** |
| Expansion connector 3V3 | CN10 | SB2 | SB1, SB13, SB23, SB25, and SB26 |
| Expansion connector 5V | CN10 | SB1 | SB2, SB13, SB17, and SB26 |
| VBAT header for 5V supply | CN3 | SB13 | SB17, SB23, and SB26 |
| VBAT header for 3V3 supply | CN3 | SB26 | SB13, SB23, and SB26 |

1. *The default configuration is shown in bold.*

*Note: Additional more complex power supply configurations might be used. In such cases, the user must carefully check solder bridge configurations according to the power tree, to avoid any damages.*

**Warning:** *Due to solder bridge usage, pay attention to SB ON and OFF configuration. Any mistake might lead to irreversible damages on the X-NUCLEO board or host.*

### 7.4.2 ST67W611M1A6B module power domains

The ST67W611M1A6B module has two supply domains: VDD_IO for I/Os and VDD33 for the radio processor unit (RPU). Those two domains can be powered in different ways: 3.3 V, 1.8 V, or directly from the battery connector (CN3), depending on the targeted application. The supply is selected with solder bridges. Table 7 shows the different power supply possibilities:

**Table 7. ST67W611M1A6B power domain capabilities**

| - | VDD_IO | VDD_RF |
|---|---|---|
| Power supplies capabilities | 1.8 and 3.3 V | VBAT and 3.3 V |

In the default MCU power supply configuration, VDDIO and VDDRF are supplied by 3V3 from the LDO (U6). To set VDD_IO to 3.3 V, SB18 must be ON and SB22 must be OFF. This is the default configuration. To set VDD_IO to 1.8 V, then SB18 must be OFF and SB22 must be ON.

*Note: When VDDIO is set to 1.8 V, all host interface signals must operate at the 1.8 V logic level. However, the ST67W611M1 CHIP_EN input requires a 3.3 V high level to activate the module. To ensure proper operation:*

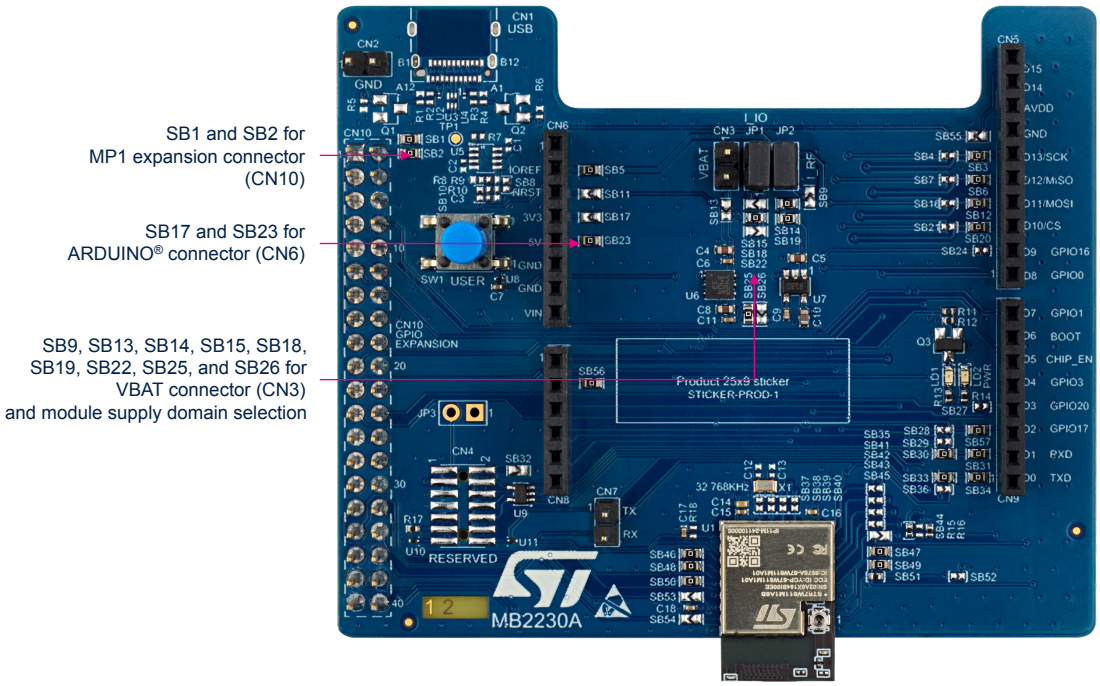
- *Confirm that the host IO driving CHIP_EN is 3.3 V tolerant.*
- *Configure the host IO driving CHIP_EN in open-drain mode. Modify the project `ioc` file or edit the `main.c` file. Replace the following line in `main.c`: `GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;` with `GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_OD;`.*
- *Connect a 33 kΩ pull-up resistor on the CHIP_EN signal to VDDRF (3.3 V). Use the R18 resistor footprint. This configuration increases current consumption by 100 µA in shutdown mode.*

**Caution:** *A hazardous solder bridge configuration can damage the board or the ST67W611M1A6B module. SB18 and SB22 are mutually exclusive: SB18 or SB22 cannot be ON at the same time. Otherwise, a voltage conflict occurs between 1V8 and 3V3 at VDD_IO.*

### 7.4.3 Power supply solder bridge locations on X-NUCLEO

Figure 9 gives the location of the power supply solder bridges on the board. The concerned solder bridges are located in three areas: GPIO selection near the CN10 connector, ARDUINO® selection near the bottom of CN6, VBAT, and power domain selection around LDOs (U6 and U7).

**Figure 9. Power supply solder bridge locations**




DT59658V1

### 7.4.4 Power supply limitations

In some configurations, the X-NUCLEO power configurations are limited:

- If the entire board is battery-powered on CN3 and the battery voltage is lower than ~3V3, it might be impossible to supply 3V3 from the LDO (U6). In this case, the only possible configuration is to supply VDDRF directly from VBAT (SB26 ON and SB25 OFF).

## 7.5 Clock sources

### 7.5.1 High-speed clock

The 40 MHz clock high precision quartz for best RF performances is internal to the ST67W611M1A6B module.

### 7.5.2 Low-speed clock (low-speed external clock)

To get the low-power mode feature with the ST67W611M1A6B module, a 32 kHz frequency clock must be provided to the ST67W611M1A6B module. This clock can be provided either:

- By the onboard 32 kHz crystal (X1). To do that, SB37 and SB38 must be ON, and SB52 remains OFF. In addition, adjust the C12 and C13 external load capacitor values if needed.
- Externally with a signal to output the 32 kHz clock on GPIO17. On the STM32U5 Nucleo-144 board (MB1549) with its STM32U575ZI microcontroller (main development stream), LSCO is on PA2. And PA2 is also on the ARDUINO® connector (A1 pin connected to GPIO17).

## 7.6 LEDs

The user LED (LD1) is at the user's disposal. This LED is 1.8 or 3.3 V-driven depending on the VDDIO power supply configuration.

#### User LED (LD1)

This blue LED is a user LED connected to ST67W611M1A6B module pin 21 (GPIO20). To light the LD1 LED, a HIGH logic state must be written in the corresponding GPIO20. There is a possibility to control LD1 from the host board through the ARDUINO® D3 pin on CN9.

When SPI is selected as the communication protocol between the host and the module, the user cannot use GPIO20 to control the user LED directly because it must be used as SPI DATA READY.

#### Power LED (LD2)

This power green LED is connected to the 3V3 of the board to indicate that the 3V3 voltage is present.

## 7.7 Push button

#### User push button (SW1)

The user button is connected to the ST67W611M1A6B module pin 1 (GPIO0). SW1 is active LOW.

## 7.8 Current consumption measurement

X-NUCLEO-67W61M1 offers the possibility to measure the ST67W611M1A6B module current consumption for both supply domains VDD_IO and VDD_RF. The measurement is done by removing a jumper and inserting an ammeter on the supply path. The current measurement method is the same for both VDD_IO and VDD_RF supplies. The two IDD jumpers are:

- JP1 (I_IO): Measures ST67W611M1A6B module I/O parts, such as GPIO and SPI.
- JP2 (I_RF): Measures ST67W611M1A6B module radio unit consumption.

The overall consumption of the ST67W611M1A6B module Wi-Fi®coprocessor is the sum of the current flowing through JP1 and JP2.

When the current measurement is not needed, JP1 and JP2 are not mounted. SB9 and SB10 must be ON to bypass the current measurement feature.

## 7.9 USART and Virtual COM port usage

The ST67W611M1A6B module USART is addressed on different connectors of the X-NUCLEO-67W61M1 board:

- On ARDUINO® (CN9) for host communication,
- On the expansion connector (CN10) for host communication,
- And on a 2-point header connector (CN7).

As the default configuration, UART is connected to the ARDUINO® connector. Solder bridges can be removed to isolate UART to a specific connector. Table 8 gives different possibilities.

**Table 8. USART usage selection**

| SB ON | SB OFF | UART available on connector |
|---|---|---|
| **SB31, SB34, SB30, and SB33**(1) | **SB29 and SB36**(1) | **ARDUINO® (CN9) and 2-pin header UART (CN7)**(1) |
| SB29 and SB36 | SB31, SB34, SB30, and SB33 | GPIO expander (CN10) only |
| SB31, SB34, SB30, SB33, SB29, and SB36 | - | ARDUINO® CN9, 2-pin header UART (CN7), and GPIO expander (CN10) |

1. *The default configuration is shown in bold.*

## 7.10 Solder bridges

Fifty-six solder bridges are located on the X-NUCLEO-67W61M1 board. All the solder bridges present in the board are used to configure several I/Os and power supply pins for feature and pinout compatibility with the host.

**Table 9. Solder bridge summary**

| Solder bridge control | Solder bridge | State(1) | Description(1) |
|---|---|---|---|
| Main 5 V input | SB2 | ON | The 5V supply is provided by the GPIO expansion connector (CN10) |
| | | **OFF** | **The 5V supply is not provided by CN10** |
| | SB13 | ON | The 5V supply is provided by the VBAT connector (CN3) |
| | | **OFF** | **The 5V supply is not provided by CN3** |
| | SB23 | **ON** | **The 5V supply is provided by the ARDUINO® connector (CN6)** |
| | | OFF | The 5V supply is not provided by CN6 |
| Main 3V3 input | SB1 | ON | The 3V3 supply is provided by the GPIO expansion connector (CN10) |
| | | **OFF** | **The 3V3 supply is not provided by CN10** |
| | SB17 | ON | The 3V3 supply is provided by the VBAT connector (CN3) |
| | | **OFF** | **The 3V3 supply is not provided by CN3** |
| | SB26 | ON | The 3V3 supply is provided by the ARDUINO® connector (CN6) |
| | | **OFF** | **The 3V3 supply is not provided by CN6** |
| SDIO signals (from ST67W611M1A6B) present at the ARDUINO® connector | SB35, SB39, SB40, SB41, SB42, SB43 | ON | The SDIOs signals are present on CN5 and CN8 (ARDUINO®) |
| | | **OFF** | **The SDIOs signals are not present on CN5 and CN8** |
| 32 KHz crystal (X1) | SB37, SB38 | ON | The 32 KHz crystal (X1) provides a 32 KHz clock to ST67W611M1A6B |
| | | **OFF** | **The 32 KHz crystal does not provide a clock to the ST67W611M1A6B** |
| VBUS detection | SB8 | ON | VBUS detection connected to 5V (USB-C® connector detection) |
| | | **OFF** | **VBUS detection not connected to 5V** |
| VBUS_DETECT range | SB10 | ON | VBUS_DETECT range is 3V3 |
| | | **OFF** | **VBUS_DETECT range is 1V8** |
| VDDIO connected at IOREF | SB5 | ON | VDDIO connected at IOREF |
| | | **OFF** | **VDDIO not connected with IOREF** |
| SPI interface connected to the Raspberry Pi® 40-pin GPIO header | SB4 | ON | SPI_CLK signal connected to the GPIO expansion connector (CN10) |
| | | **OFF** | **SPI_CLK signal not connected to CN10** |
| | SB7 | ON | SPI_MISO signal connected to the GPIO expansion connector (CN10) |
| | | **OFF** | **SPI_MISO signal not connected to CN10** |
| | SB16 | ON | SPI_MOSI signal connected to theGPIO expansion connector (CN10) |
| | | **OFF** | **SPI_MOSI signal not connected to CN10** |
| | SB21 | ON | SPI_NSS signal connected to the GPIO expansion connector (CN10) |
| | | **OFF** | **SPI_NSS signal not connected to CN10** |

| Solder bridge control | Solder bridge | State(1) | Description(1) |
|---|---|---|---|
| UART interface connected to the GPIO expansion connector | SB29 | ON | UART_RXD signal connected to theGPIO expansion connector (CN10) |
| | | **OFF** | **UART_RXD signal not connected to CN10** |
| | SB36 | ON | UART_TXD signal connected to the GPIO expansion connector (CN10) |
| | | **OFF** | **UART_TXD signal not connected to CN10** |
| GPIO signals connected to the GPIO expansion connector | SB24 | ON | GPIO16 signal connected to the GPIO expansion connector (CN10) |
| | | **OFF** | **GPIO16 signal not connected to CN10** |
| | SB27 | ON | GPIO20 signal connected to the GPIO expansion connector (CN10) |
| | | **OFF** | **GPIO20 signal not connected to CN10** |
| | SB28 | ON | GPIO17 signal connected to the GPIO expansion connector (CN10) |
| | | **OFF** | **GPIO17 signal not connected to CN10** |
| CHIP_EN on NRST pin of ARDUINO® connector | SB11 | ON | CHIP_EN signal connected to NRST pin of ARDUINO® connector |
| | | **OFF** | **CHIP_EN signal not connected to NRST pin of ARDUINO® connector** |
| GPIO17 on CN8 ARDUINO® connector | SB56 | **ON** | **GPIO17 connected on CN8 ARDUINO® connector (ARD.A2)** |
| | | OFF | GPIO17 not connected on CN8 ARDUINO® connector (ARD.A2) |
| SPI interface connected to CN5 (ARDUINO® connector) | SB3 | ON | SPI_CLK signal connected to (CN5) ARDUINO® connector |
| | | **OFF** | **SPI_CLK signal not connected to CN5** |
| | SB6 | ON | SPI_MISO signal connected to (CN5) ARDUINO® connector |
| | | **OFF** | **SPI_MISO signal not connected to CN5** |
| | SB12 | ON | SPI_MOSI signal connected to (CN5) ARDUINO® connector |
| | | **OFF** | **SPI_MOSI signal not connected to CN5** |
| | SB20 | ON | SPI_NSS signal connected to (CN5) GPIO ARDUINO® connector |
| | | **OFF** | **SPI_NSS signal not connected to CN5** |
| UART interface connected to CN5 (ARDUINO® connector) | SB31 | **ON** | **UART_RXD signal connected to (CN9) ARDUINO® connector ARD.D1** |
| | | OFF | UART_RXD signal not connected to (CN9) ARDUINO® connector ARD.D1 |
| | SB34 | **ON** | **UART_TXD signal connected to (CN9) ARDUINO® connector ARD.D0** |
| | | OFF | UART_TXD signal not connected to (CN9) ARDUINO® connector ARD.D0 |
| UART interface connected to CN7 (2-pin header) | SB30 | **ON** | **UART_RXD signal connected to (CN7)** |
| | | OFF | UART_RXD signal not connected to (CN7) |
| | SB33 | **ON** | **UART_TXD signal connected to (CN7)** |
| | | OFF | UART_TXD signal not connected to (CN7) |
| USB pins connected to USB connector (CN1) | SB46, SB48 | ON | USB pins (USB_DM and USB_DP) connected to USB connector (CN1) |
| | | **OFF** | **USB pins not connected to USB connector (CN1)** |
| SB to short circuit JP2 | SB9 | ON | JP2 is shorted (no I_RF current measurement is possible) |

| Solder bridge control | Solder bridge | State(1) | Description(1) |
|---|---|---|---|
| SB to short circuit JP2 | SB9 | **OFF** | **JP2 is open (I_RF current measurement is possible at JP2)** |
| SB to short circuit JP1 | SB15 | ON | JP1 is shorted (no I_IO current measurement is possible) |
| | | **OFF** | **JP2 is open (I_IO current measurement is possible at JP1)** |
| 3V3 connected to JP2 | SB14 | **ON** | **3V3 voltage connected to JP2** |
| | | OFF | 3V3 voltage not connected to JP2 |
| 3V3 LDO input connected to 3V3 | SB19 | **ON** | **The 3V3 LDO (U7) input is connected to 3V3** |
| | | OFF | The 3V3 LDO (U7) input is isolated from 3V3 (U7 is not powered) |
| 5V LDO output connected to 3V3 | SB25 | **ON** | **The 5V LDO (U6) output is connected to 3V3** |
| | | OFF | The 5V LDO (U6) output is isolated from 3V3 |
| CN4 pin2 connected to GND | SB32 | ON | The CN4 pin2 is connected to GND |
| | | **OFF** | **The CN4 pin2 is isolated from GND** |
| VDDIO voltage at either 1V8 or 3V3 | SB18 | **ON** | **VDDIO = 3V3 (SB18 and SB22 are exclusive)** |
| | | OFF | VDDIO = 1V8 (SB18 and SB22 are exclusive) |
| | SB22 | ON | VDDIO = 1V8 (SB18 and SB22 are exclusive) |
| | | **OFF** | **VDDIO = 3V3 (SB18 and SB22 are exclusive)** |
| GPIO2/BOOT_SEL connected to T_JTDO | SB44 | **ON** | **GPIO2/BOOT_SEL connected to T_JTDO** |
| | | OFF | GPIO2/BOOT_SEL not connected to T_JTDO |
| GPIO3 connected to T_JTDI | SB45 | ON | GPIO3 connected to T_JTDI |
| | | **OFF** | **GPIO3 not connected to T_JTDI** |
| GPIO1 connected to T_JTCK | SB47 | **ON** | **GPIO1 connected to T_JTCK** |
| | | OFF | GPIO1 not connected to T_JTCK |
| GPIO0 connected to T_JTMS | SB49 | **ON** | **GPIO0 connected to T_JTMS** |
| | | OFF | GPIO0 not connected to T_JTMS |
| GPIO20 connected to LED1 | SB50 | **ON** | **GPIO20 connected to LED1** |
| | | OFF | GPIO20 not connected to LED1 |
| GPIO0 connected to the button | SB51 | **ON** | **GPIO0 connected to the button** |
| | | OFF | GPIO0 not connected to the button |
| GPIO16 connected to the button | SB52 | ON | GPIO16 connected to the button |
| | | **OFF** | **GPIO16 not connected to the button** |
| GPIO30 connected to T_JTDO | SB53 | ON | GPIO30 connected to T_JTDO |
| | | **OFF** | **GPIO30 not connected to T_JTDO** |
| VBUS_DETECT connected to GPIO28 | SB54 | ON | VBUS_DETECT connected to GPIO28 |
| | | **OFF** | **VBUS_DETECT not connected to GPIO28** |
| VBUS_DETECT connected to ARDUINO® connector | SB55 | ON | VBUS_DETECT connected to ARD.D15 (ARDUINO® connector CN5) |
| | | **OFF** | **VBUS_DETECT not connected to ARD.D15** |

1. *The default configuration is shown in bold.*

# 8 Board connectors

The four different board connectors and their pinouts are detailed in this section.

## 8.1 MB2230 2-pin VBAT connector (CN3)

The CN3 connector is a standard 2-pin VBAT connector (2.54-mm pitch). The X-NUCLEO board can be powered from the VBAT connector (CN3) by either:

- Plugging an external 5 V voltage supply (from 3.6 up to 5.5 V)
- Plugging a battery pack

**Figure 10. 2-pin VBAT connector (CN3)**

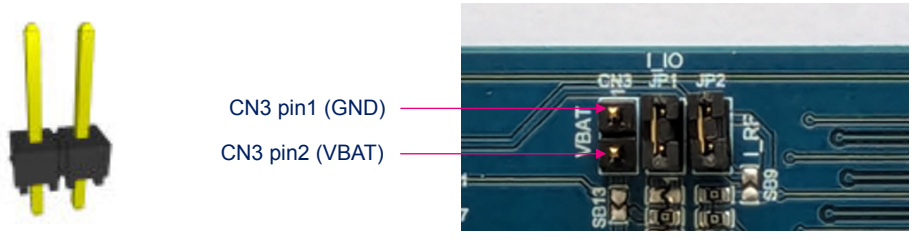



**Table 10. External power source VBAT (CN3)**

| Connector | Pin number | Pin name | Signal name | Function |
|---|---|---|---|---|
| CN3 | 1 | GND | GND | GND |
| | 2 | VBAT | VBAT | VBAT power |

## 8.2 MB2230 2-pin UART connector (CN7)

The CN7 connector is a standard 2-pin UART connector (2.54-mm pitch). CN7 is placed on the X-NUCLEO to easily probe UART signals.

**Figure 11. 2-pin UART connector (CN3)**

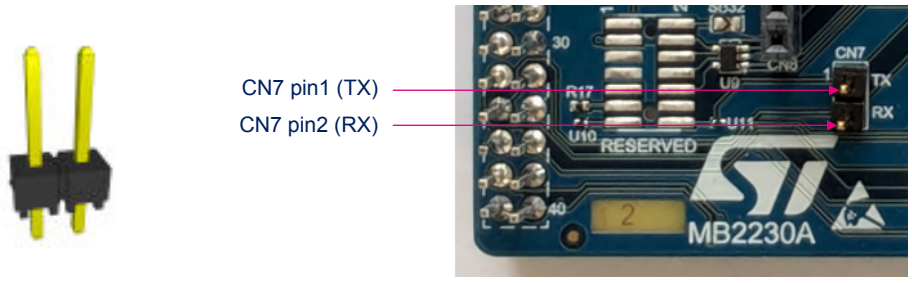


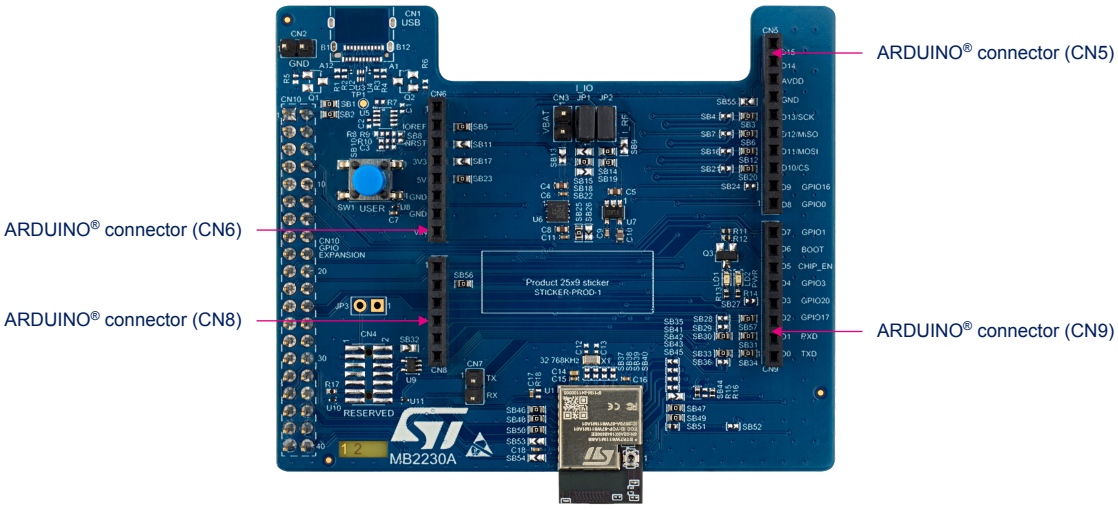

**Table 11. UART connector (CN7)**

| Connector | Pin number | Pin name | Signal name | Function |
|---|---|---|---|---|
| CN7 | 1 | TX | TX | UART TX (ST67W611M1 module side)<br>UART RX (host side) |
| | 2 | RX | RX | UART TX (ST67W611M1 module side)<br>UART RX (host side) |

## 8.3 ARDUINO® Uno V3 connector

CN5, CN6, CN8, and CN9 are bottom-male, top-female connectors compatible with the ARDUINO® standard. These connectors can be used for SPI data exchange with the host, power supply, USART, LED, button, and GPIO control. Refer to Figure 12 for their location.

**Figure 12. ARDUINO® connectors**



The pinout for the ARDUINO® connector is listed in Table 12.

**Table 12. ARDUINO® connector pinout**

| Connector | Pin number | Pin name | Signal name | ST67W611M1A6B module pin(1) | Function(1) |
|---|---|---|---|---|---|
| CN6 | 1 | NC | - | - | - |
| | 2 | IOREF | - | - | I/O reference |
| | 3 | NRST | T_NRST | NRST | RESET |
| | 4 | 3V3 | - | - | 3.3 V input |
| | 5 | 5V | - | - | 5 V output |
| | 6 | GND | - | - | GND |
| | 7 | GND | - | - | GND |
| | 8 | VIN | - | - | VIN to host Board |
| CN8 | 1 | - | GPIO10 | GPIO10 | SDIO.D2 |
| | 2 | - | GPIO17 | GPIO17 | External 32 KHz |
| | 3 | - | GPIO12 | GPIO12 | SDIO.CMD |
| | 4 | - | GPIO13 | GPIO13 | SDIO.CLK |
| | 5 | - | GPIO14 | GPIO14 | SDIO.D0 |
| | 6 | - | GPIO15 | GPIO15 | SDIO.D1 |
| CN5 | 10 | D15 | - | - | VBUS_DETECT |
| | 9 | D14 | GPIO11 | GPIO11 | SDIO.D3 |
| | 8 | AVDD | - | - | - |
| | 7 | GND | - | - | GND |
| | 6 | D13/SCK | SPI1_CLK | GPIO29 | SPI1_SCK |
| | 5 | D12/MISO | SPI1_MISO | GPIO30 | SPI1_MISO |
| | 4 | D11/MOSI | SPI1_MOSI | GPIO27 | SPI1_MOSI |
| | 3 | D10/CS | SP1/NSS | GPIO28 | SPI1_NSS |
| | 2 | D9 | GPIO16 | GPIO16 | **GPIO16**/button |
| | 1 | D8 | GPIO0 | GPIO0 | GPIO0/**button** |
| CN9 | 8 | D7 | GPIO1 | GPIO1 | T_JTCK |
| | 7 | D6 | GPIO2/BOOT_SEL | GPIO2/BOOT_SEL | **BOOT**/T_JTDO |
| | 6 | D5 | CHIP_EN | CHIP_EN | CHIP_EN |
| | 5 | D4 | GPIO3 | GPIO3 | T_JTDI |
| | 4 | D3 | GPIO20 | GPIO20 | LED1 |
| | 3 | D2 | GPIO17 | GPIO17 | GPIO17 |
| | 2 | D1/TX | UART_RXD | UART_RXD | UART_TX |
| | 1 | D0/RX | UART_TXD | UART_TXD | UART_RX |

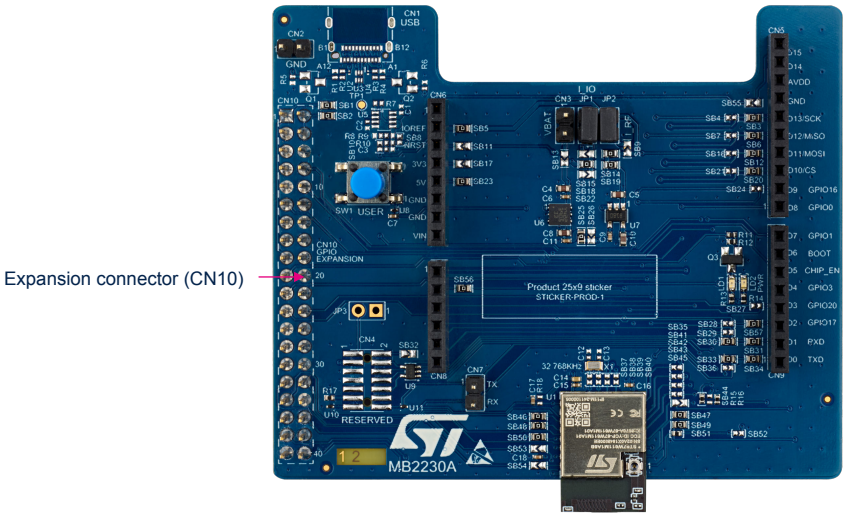
1. *The default configuration is in bold.*

## 8.4 Raspberry Pi® GPIO header

CN10 is a female bottom, male top standard expansion connector, compatible with the Raspberry Pi® GPIO header. It transmits SPI, USART, power supply, and control signals. It is intended to interface X-NUCLEO with STM32MPxx DK1 or any other boards with an expansion connector. Refer to Table 13 for their location.

CN5, CN6, CN8, and CN9 are bottom-male, top-female connectors compatible with the ARDUINO® standard. These connectors can be used for SPI data exchange with the host, power supply, USART, LED, button, and GPIO control. Refer to Figure 13 for their location.

**Figure 13. X-NUCLEO GPIO expansion connector**




DT59664V1

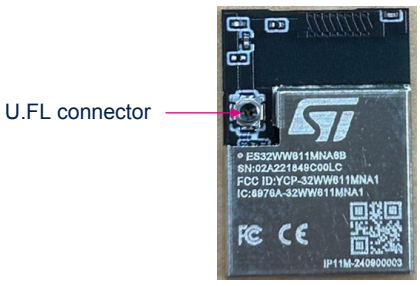
Table 13. **Expansion connector (CN10) pinout**

| GPIO | | | |
|---|---|---|---|
| **Pin** | **Signal** | **Pin** | **Signal** |
| 1 | 3V3 | 2 | 5V |
| 3 | - | 4 | 5V |
| 5 | - | 6 | GND |
| 7 | - | 8 | UART_RXD |
| 9 | GND | 10 | UART_TXD |
| 11 | GPIO0 | 12 | - |
| 13 | - | 14 | GND |
| 15 | - | 16 | - |
| 17 | 3V3 | 18 | - |
| 19 | SPI1_MOSI | 20 | GND |
| 21 | SPI1_MISO | 22 | - |
| 23 | SPI1_CLK | 24 | SPI1_NSS |
| 25 | GND | 26 | - |
| 27 | - | 28 | - |
| 29 | - | 30 | GND |
| 31 | GPIO2/BOOT_SEL | 32 | GPIO3 |
| 33 | CHIP_EN | 34 | GND |
| 35 | GPIO16 | 36 | GPIO1 |
| 37 | GPIO17 | 38 | - |
| 39 | GND | 40 | GPIO20 |

## 8.5 RF connector

The ST67W611M1A6B module includes the Wi-Fi®/Bluetooth®/802.15.4 transceiver, the complete front-end structure, a PCB antenna (12.28 x 17.28 x 2.2 mm, 1.27-mm pitch, 32-pin, 4-side LGA (4-layer PCB)). An RF U.FL connector is also implemented on the module to allow conducted measurements.

Figure 14. **U.FL connector**




To carry out conducted measurements on the ST67W611M1A6B module, a specific cable connector such as the Murata MXHQ87WJ3000, available from authorized distributors, is required: https://www.mouser.fr/ProductDetail/Murata-Electronics/MXHQ87WJ3000?qs=tCuqrAZu8bYUL83%2Fv%2FTPMQ%3D%3D

# 9 ST67W611M1A6B module I/O assignment

Table 14. **ST67W611M1A6B module I/O assignment**

| Pin | Pin name | Power domain | ARDUINO® pin | Main/optional feature(1) |
|---|---|---|---|---|
| 1 | RESERVED | VDDIO | ARD_D8 | Button |
| 2 | RESERVED | VDDIO | ARD_D7 | - |
| 3 | BOOT | VDDIO | ARD_D6 | BOOT |
| 4 | RESERVED | VDDIO | ARD_D4 | - |
| 5 | RESERVED | VDDIO | ARD_A3 | - |
| 6 | RESERVED | VDDIO | ARD_D14 | - |
| 7 | RESERVED | VDDIO | ARD_A1 | - |
| 8 | RESERVED | VDDIO | ARD_A5 | - |
| 9 | VDDIO | - | - | Host I/O voltage input |
| 10 | GND | GND | ARD_GND | GND |
| 11 | RESERVED | VDDIO | ARD_A4 | - |
| 12 | RESERVED | VDDIO | ARD_A6 | - |
| 13 | XTAL32K_OUT | VDD33 | ARD_A2/ARD_D2 | 32.768 kHz clock is optional. 32.768 kHz coming from the host processor can be provided on this pin. |
| 14 | XTAL32K_IN | VDD33 | ARD_D9 | 32.768 kHz quartz is optional. |
| 15 | GND | GND | ARD_GND | GND |
| 16 | VDD33 | - | ARD_3V3 | Power input (3.3 V) |
| 17 | CHIP_EN | - | ARD_D5 | Chip power ON |
| 18 | GND | GND | ARD_GND | GND |
| 19 | RESERVED | - | - | - |
| 20 | RESERVED | - | - | - |
| 21 | SPI_RDY | VDDIO | ARD_D3 | Data ready to be transmitted to host processor/user LED (LD1)(2) |
| 22 | UART_TX | VDDIO | ARD_D0 | UART_TXD (module side) |
| 23 | UART_RX | VDDIO | ARD_D1 | UART_RXD (module side) |
| 24 | SPI_CS | VDDIO | ARD_D10 | SPI_NSS |
| 25 | VDDIO | - | - | Host I/O voltage input |
| 26 | GND | GND | ARD_GND | GND |
| 27 | SPI_MOSI | VDDIO | ARD_D11 | Input SPI data from the master/host to the module |
| 28 | SPI_CLK | VDDIO | ARD_D13 | Input of SPI clock signal generated by the SPI master/host |
| 29 | SPI_MISO | VDDIO | ARD_D12 | Output of SPI data from the module to the SPI master/host |
| 30 | GND | GND | ARD_GND | GND |
| 31 | NC | - | - | Pin not connected |
| 32 | GND | GND | ARD_GND | GND |
| 33 | GND | GND | ARD_GND | Expose pad |
| 34 | GND | GND | ARD_GND | Expose pad |
| 35 | GND | GND | ARD_GND | Expose pad |
| 36 | GND | GND | ARD_GND | Expose pad |

1. *The default configuration is shown in bold.*

2. *When SPI is the communication protocol between the host and the module, the user cannot use SPI_RDY to control the user LED directly because it must be used as SPI DATA READY.*

# 10 X-NUCLEO-67W61M1 product information

## 10.1 Product marking

The product and each board composing the product are identified with one or several stickers. The stickers, located on the top or bottom side of each PCB, provide product information:

- Main board featuring the target device: product order code, product identification, serial number, and board reference with revision.

  Single-sticker example:

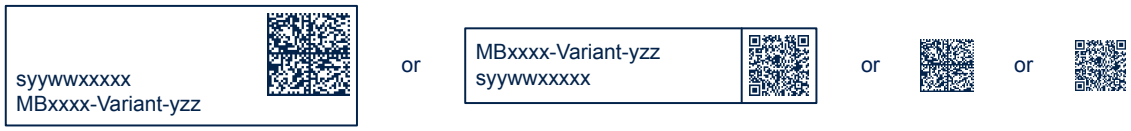
  
  

  Dual-sticker example:

- Other boards if any: board reference with revision and serial number.

  Examples:

On the main board sticker, the first line provides the product order code, and the second line the product identification.

On all board stickers, the line formatted as *"MBxxxx-Variant-yzz"* shows the board reference *"MBxxxx"*, the mounting variant *"Variant"* when several exist (optional), the PCB revision *"y"*, and the assembly revision *"zz"*, for example B01. The other line shows the board serial number used for traceability.

Products and parts labeled as *"ES"* or *"E"* are not yet qualified or feature devices that are not yet qualified. STMicroelectronics disclaims any responsibility for consequences arising from their use. Under no circumstances will STMicroelectronics be liable for the customer's use of these engineering samples. Before deciding to use these engineering samples for qualification activities, contact STMicroelectronics' quality department.

*"ES"* or *"E"* marking examples of location:

- On the targeted STM32 that is soldered on the board (for an illustration of STM32 marking, refer to the STM32 datasheet *Package information* paragraph at the *www.st.com* website).
- Next to the ordering part number of the evaluation tool that is stuck, or silk-screen printed on the board.

Some boards feature a specific STM32 device version, which allows the operation of any bundled commercial stack/library available. This STM32 device shows a *"U"* marking option at the end of the standard part number and is not available for sales.

To use the same commercial stack in their applications, the developers might need to purchase a part number specific to this stack/library. The price of those part numbers includes the stack/library royalties.

## 10.2 X-NUCLEO-67W61M1 product history

**Table 15. Product history**

| Order code | Product identification | Product details | Product change description | Product limitations |
|---|---|---|---|---|
| X-NUCLEO-67W61M1 | XN67W61M1$CZ1 | MCU:<br>• ST67W611M1A6B | Initial revision | The current module revision is at 100 µA power consumption in shutdown mode (instead of 0.2 µA). |
| | | MCU errata sheet:<br>• No errata sheet. | | |
| | | Board:<br>• MB2230-ANTENNA-A04 (expansion board) | | |
| | XN67W61M1$CZ2 | MCU:<br>• ST67W611M1A6B | Box product and certification label sticker | The current module revision is at 100 µA power consumption in shutdown mode (instead of 0.2 µA). |
| | | MCU errata sheet:<br>• No errata sheet. | | |
| | | Board:<br>• MB2230-ANTENNA-A04 (expansion board) | | |
| | XN67W61M1$CZ3 | MCU:<br>• ST67W611M1A6B | Japanese MIC certification | No limitation |
| | | MCU errata sheet:<br>• No errata sheet. | | |
| | | Board:<br>• MB2230-67W611-B01 (expansion board) | | |

## 10.3 Board revision history

**Table 16. Board revision history**

| Board reference | Board variant and revision | Board change description | Board limitations |
|---|---|---|---|
| MB2230 (expansion board) | ANTENNA-A04 | Initial revision | No limitation |
| | 67W611-B01 | PCB change (PCB rev B) to align silkscreen with pin names of the ST67W611MNA6B module. No BOM change. | No limitation |

# 11 Compliance statements and conformity declarations

## 11.1 Federal Communications Commission (FCC) compliance statement

**Product identification**

- Identification of the product: X-NUCLEO-67W61M1
  Contains a transmitter/receiver module from STMicroelectronics: ST67W611M1
  FCC ID: YCP-67W611M1A01

**Part 15.19**

This device complies with part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

**Part 15.21**

Any changes or modifications to this equipment not expressly approved by STMicroelectronics may cause harmful interference and void the user's authority to operate this equipment.

**Part 15.105**

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

*Note: Use only shielded cables.*

To satisfy FCC RF exposure requirements, a separation distance of 20 cm or more should be maintained between the antenna of this device and persons during operation. To ensure compliance, operation at a closer distance than this is not recommended. This transmitter must not be collocated or operating in conjunction with any other antenna or transmitter.

**Responsible Party - U.S. Contact Information:**

Francesco Doddo
STMicroelectronics, Inc.
200 Summit Drive | Suite 405 | Burlington, MA 01803
USA
Telephone: +1 781-472-9634

## 11.2 Innovation, Science and Economic Development Canada (ISED) compliance statement

**Product identification**

- Identification of the product: X-NUCLEO-67W61M1
  Contains a transmitter/receiver module from STMicroelectronics: ST67W611M1
  IC: 8976A-67W611M1A01

**Compliance statement**

Notice: This device complies with ISED Canada licence-exempt RSS standard(s). Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

This device complies with ISED Canada RF radiation exposure limits set forth for general population for mobile application (uncontrolled exposure).

To satisfy ISED RF exposure requirements, a separation distance of 20 cm or more should be maintained between the antenna of this device and persons during operation. To ensure compliance, operation at a closer distance than this is not recommended. This transmitter must not be collocated or operating in conjunction with any other antenna or transmitter.

**This product complies with the ICES-003 standard class B of the ISED regulation.**

ISED Canada ICES-003 Compliance Label: CAN ICES (B)/NMB (B).

*Note: Use only shielded cables.*

**Identification du produit**

- Identification du produit : X-NUCLEO-67W61M1
  Contient un module émetteur/récepteur de STMicroelectronics : ST67W611M1
  IC : 8976A-67W611M1A01

**Déclaration de conformité**

Avis : Le présent appareil est conforme aux CNR d'ISDE Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes : (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

Cet appareil est conforme aux limites d'exposition aux rayonnements RF d'ISDE Canada établies pour la population générale pour les applications mobiles (exposition non contrôlée).

Pour satisfaire aux exigences d'ISDE en matière d'exposition aux RF, une distance de séparation de 20 cm ou plus doit être maintenue entre l'antenne de cet appareil et les personnes pendant son fonctionnement. Pour garantir la conformité, il n'est pas recommandé de l'utiliser à une distance plus proche que celle-ci. Cet appareil ne doit pas être placé à proximité ou fonctionner en conjonction avec une autre antenne ou un autre émetteur.

**Ce produit est conforme à la norme NMB-003 classe B de la ISDE.**

Étiquette de conformité à la NMB-003 d'ISDE Canada : CAN ICES (B) / NMB (B).

*Note: Utiliser uniquement des câbles blindés.*

## 11.3 Japanese Radio Certification: Ministry of Internal affairs and Communications (MIC) conformity

**Japan MIC conformity**

Identification of the product: X-NUCLEO-67W61M1

This product is compliant with the Japanese radio law (MIC).

Contains a transmitter/receiver module from STMicroelectronics: ST67W611M1

- TELEC certification: 020-250100
- JATE certification: 250063020

## 11.4 UKCA conformity

**Simplified UK declaration of conformity**

Hereby, the manufacturer STMicroelectronics, declares that the radio equipment type X-NUCLEO-67W61M1 is in compliance with the UK Radio Equipment Regulations 2017 (UK S.I. 2017 No. 1206) and with the Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 (UK SI 2012 No. 3032).

The full text of the UK declaration of conformity is available at the following internet address: *www.st.com*.

*Note: Use only shielded cables.*

## 11.5 CE conformity

### 11.5.1 Simplified EU declaration of conformity

Hereby, STMicroelectronics declares that the radio equipment type X-NUCLEO-67W61M1 is in compliance with directives 2011/53/EU and 2015/863/EU (RoHS), and 2014/53/EU (RED).

Frequency range used in transmission and maximal radiated power in this range for X-NUCLEO-67W61M1:

- Frequency range: 2400 - 2483.5 MHz (Wi-Fi®/Bluetooth®)
- Maximal power for Wi-Fi®: 100 mW E.I.R.P.
- Maximal power for Bluetooth® LE: 10 mW E.I.R.P.

The full text of the EU declaration of conformity is available on demand at the following internet address: *www.st.com*.

*Note:*
- *RoHS: Restriction of hazardous substances*
- *RED: Radio equipment directive*

*Note: Use only shielded cables.*

### 11.5.2 Déclaration de conformité UE simplifiée

STMicroelectronics déclare que l'équipement radioélectrique du type X-NUCLEO-67W61M1 est conforme aux directives 2011/53/UE et 2015/863/UE (LdSD), et à la directive 2014/53/UE (RED).

Plage de fréquences utilisée en transmission et puissance rayonnée maximale dans cette plage pour X-NUCLEO-67W61M1 :

- Plage de fréquences : 2400 - 2483,5 MHz (Bluetooth®)
- Puissance maximale Wi-Fi® : 100 mW E.I.R.P.
- Puissance maximale Bluetooth® LE : 10 mW E.I.R.P.

Le texte complet de la déclaration UE de conformité est disponible sur demande à l'adresse internet suivante : *www.st.com*.

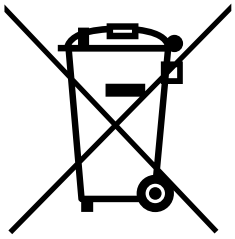
*Note:*
- *LdSD : directive sur la limitation de l'utilisation des substances dangereuses*
- *RED : directive sur les équipements radio-électriques*

*Note: Utiliser uniquement des câbles blindés.*

# 12 Product disposal

**Disposal of this product: WEEE (Waste Electrical and Electronic Equipment)**

(Applicable in Europe)



This symbol on the product, accessories, or accompanying documents indicates that the product and its electronic accessories must not be disposed of with household waste at the end of their working life.

To prevent possible harm to the environment and human health from uncontrolled waste disposal, separate these items from other types of waste and recycle them responsibly at a designated collection point to promote the sustainable reuse of material resources.

Household users:

Contact the retailer that you purchased the product from or your local authority for details of your nearest designated collection point.

Business users:

Contact your dealer or supplier for further information.

# Revision history

Table 17. **Document revision history**

| Date | Revision | Changes |
|---|---|---|
| 02-Jun-2025 | 1 | Initial release. |
| 29-Apr-2026 | 2 | Updated:<br>• Document title<br>• Introduction<br>• Section 6.1: Getting started<br>• Figure 3. X-NUCLEO-67W61M1 hardware block diagram<br>• Figure 4. X-NUCLEO-67W61M1 top layout<br>• Table 5. X-NUCLEO host board connections<br>• Section 7.4.1: Board power supply<br>• Section 7.4.2: ST67W611M1A6B module power domains<br>• Section 7.5: Clock sources<br>• Table 14. ST67W611M1A6B module I/O assignment<br>• Section 10.2: X-NUCLEO-67W61M1 product history<br>• Section 10.3: Board revision history<br>• Section 11: Compliance statements and conformity declarations<br>Added:<br>• Section 5.3: Delivery recommendations<br>• Section 5.4: Power supply<br>• Section 11.3: Japanese Radio Certification: Ministry of Internal affairs and Communications (MIC) conformity |

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